

CONTACT

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🎓 Nima Darabi

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REFERENCES

Francesco Viti

Associate Professor
University of Luxembourg

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Guido Gentile

Associate Professor
Sapienza University of Rome

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Siamak Talatahari

Associate Professor
Macquarie University

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RESEARCH INTERESTS

- Transportation Planning
- Sustainable Transportation
- Physical Modelling
- Data Analysis
- Optimization

SKILLS

Machine Learning 2+ yrs

Python 3+ yrs

MATLAB 5+ yrs

QGIS 2+ yrs

TransCAD 1+ yrs

MATSim 1+ yrs

NIMA DARABI

Graduate Researcher

EDUCATION

M.Sc. - Transportation System Engineering
Sapienza, Rome (Italy)

2021 - 2023

Passed with GPA: **110 cum laude/110**

Thesis title: POTENTIALS of GOOGLE POPULAR TIMES DATA In MOBILITY and TRANSPORTATION ANALYSIS

B.Sc. - Civil Engineering
University Of Tabriz - Tabriz (Iran)

2015 - 2019

Passed with GPA: **17.60/20**

PUBLICATIONS

The fusion-fission optimization (FuFiO) algorithm
Scientific Reports (ISSN: 2045-2322) Vol 12, Issue 1, 2022

Status: Published

DOI: <https://doi.org/10.1038/s41598-022-16498-4>

Assessing Risk Acceptability and Tolerability in Italian Tunnels with the Quantum Gu@larp Model
Entropy (ISSN 1099-4300), Entropy 26, no. 1: 40

Status: Published

DOI: <https://doi.org/10.3390/e26010040>

How Local Businesses Shape Train Stations Demand in Luxembourg: A Data-Driven Approach
MT-ITS 2025 Conference

Status: Accepted (To be Published)

Bus Station Demand Prediction Using Crowdsourced Data and Deep Learning Models
BIVEC-GIBET 2025 Conference

Status: Accepted (To be Published)

FuFiO algorithm for Engineering problems

Status: Working paper

ACHIEVEMENTS

Duolingo English Test

135 out of 160

Issued on 2023

The Duolingo English Test (DET) certification, a product of Duolingo, attests to the English language proficiency of non-native speakers

HOBBIES

- Listening to music
- Playing the piano
- Reading literature
- Walking in nature
- Playing Chess

PROJECTS

Freight Transport and Logistics

2022

Tool: TransCAD

- Cleaned and analyzed city data in Italy to identify destination points.
- Optimized distribution planning of goods using traffic assignment techniques and shortest path algorithms.
- Developed and implemented plans to address sustainability challenges and improve the city's environmental impact.

Transport Policies

2022

Tool: TransCAD

- Analyzed data and characteristics of streets in an Italian city.
- Identified sustainability issues and developed solutions to address them.
- Implemented plans to reduce problems and improve sustainability.

Urban and Regional Policy

2022

Tool: Excel

- Scrutinized the development of sustainable urban mobility plans (SUMP) for Tehran, Iran's capital city.

Traffic Engineering and ITS

2022

Tool: Python and MATLAB

- Analyzed intersection data in Rome, and calculated their characteristics based on Highway Capacity Manual.
- Optimized traffic lights using conventional and metaheuristic methods to reduce waiting time.
- Studied the maximum green wave bandwidth problem.

Safety and Risk Analysis

2022

Tool: Python and MATLAB

- Demonstrated understanding of the ALARP principle and its application as a risk management framework.
- Delved deeper into the meanings of key risk indicators, such as Risk Quantum, Individual Risk, Societal Risk, and Expected Annual Damage.
- Utilized these indicators to assess risk exposure levels and promote safety in road tunnels.

WORKSHOPS & CONFERENCES

5th National Conference on Distributed Computing and Big Data Processing

NOV 2019

Tabriz, Azarbaijan Shahid Madani University, Iran

Workshop on Data Science and Deep Learning with Python

NOV 2019

Tabriz, Azarbaijan Shahid Madani University, Iran

HONORS & AWARDS

- Scholarship in Bachelor of Science
- Scholarship in Master of Science
- Thesis abroad scholarship from the University of Sapienza to accomplish master's thesis